## Transition tables

D - Flip-Flop

| present | next | input |
|---|---|---|
| 0 → | 0 | 0 |
| 0 → | 1 | 1 |
| 1 → | 0 | 0 |
| 1 → | 1 | 1 |

JK - Flip-Flop

| Present | next | Input J | Input K |
|---|---|---|---|
| 0 → | 0 | 0 | x |
| 0 → | 1 | 1 | x |
| 1 → | 0 | x | 1 |
| 1 → | 1 | x | 0 |

## Building a K-map

1 - State diagram of output sequence

2 - State table of output sequence

3 - transition table for F-F we want to use

- Present & next outputs
- necessary inputs to create next outputs

| Present $Q_C Q_B Q_A$ | next $Q_C Q_B Q_A$ | input $D_C D_B D_A$ |
|---|---|---|
| 0 0 0 | 0 1 0 | 0 1 0 |
| 0 1 0 | 1 1 0 | 1 1 0 |
| 1 1 0 | 0 0 1 | 0 0 1 |
| 0 0 1 | 0 0 0 | 0 0 0 |

$Q_A$

| $Q_C Q_B$ | 0 | 1 |
|---|---|---|
| 0 0 | 0 | 0 |
| 0 1 | 1 | x |
| 1 1 | 0 | x |
| 1 0 | x | x |

$Q_A$

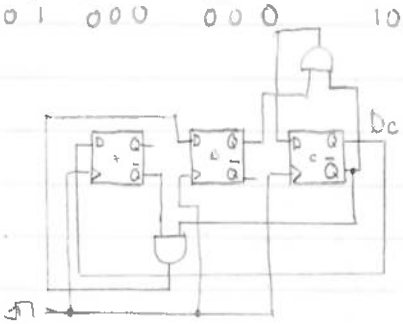
| $Q_C Q_B$ | 0 | 1 |
|---|---|---|
| 0 0 | 0 | 1 |
| 0 1 | 1 | x |
| 1 1 | 0 | x |
| 1 0 | x | x |

$Q_A$

| $Q_C Q_B$ | 0 | 1 |
|---|---|---|
| 0 0 | 0 | 0 |
| 0 1 | 0 | 1 |
| 1 1 | 1 | x |
| 1 0 | x | x |

$D_C$ $D_B$ $D_A$

$D_C = Q_B Q_C'$ $D_L = Q_A' Q_C'$ $D_A = Q_C$



## Binary synchronous counters

### Binary counter with JK flip-flop

- Build state diagram
- Build state table consisting of
  - current state output
  - next state output
  - JK inputs for each flip-flop
- Use K-maps to simplify expressions for JK inputs.
- Build the circuit for the counter

Ex:

0 → 1 → 2 → 3 → 4 → 5 → 6 → 7

3-bit binary counter

- 3 JK flip-flops needed
- Current state & next state outputs are 3 bits each
- 3 pairs of JK inputs

all states possible

### Synchronous BCD

- 4 bits for each number

0000 → 0001 → 0010 → 0011 → ...

Same design process as others

Ex:

0 → 3 → 5 → 7 → 6 → 0

- Same design process as before
- for missing states use don't cares.

don't cares for the next state & inputs

### Ring counter

- trains serial output of the last Flip-Flop of a shift register & provides it to the serial input of the first flip-flop.
- a.k.a. a re-circulating shift register.

### Johnson counter

does the same as ring counter but it uses the complement output of the last output (D-Flip-Flop).